# Project Report Compliance Review

https://gemini.google.com/app/688cd4577ef4e02b

*User prompt: Me and my course colleage have this report answering uploaded project. Is there anything obvious that we are missing from the task description?*

Response: Comparing **NLP_halla (4).pdf** against the project requirements in **Multilingual_machine_reading____DIKU_NLP_2026_2027_project-v4 (3).pdf** reveals two critical compliance issues, several specific task omissions, and an obvious caption typo:

## 1. Critical Compliance & Grading Risks

- **Total Absence of References / Bibliography:**

  There is **no References section anywhere in the paper**, and zero bibliographic citations (`\cite{...}`) appear in the body text. The report references benchmarks (TyDi QA, XOR QA, SQuAD v2), pretrained models (XLM-RoBERTa, NLLB-200), and packages without citations. The task specification explicitly states:

  > *"References, a reproducibility appendix, the contribution statement and AI declaration do not count towards the limit."* (p. 2)
  >
  > *"Acknowledge lecture/lab code, external implementations, models, datasets, translation systems and other sources."* (p. 7)
  >
  > Submitting without a reference list directly impacts the 5% communication grade and risks academic integrity penalties.

- **GenAI Declaration Template & Admitted Incomplete Logs:**

  In Appendix B (p. 7), the declaration is written as custom prose rather than using the **mandatory University of Copenhagen standard template** linked in footnote 2 (p. 1). More critically, the text states:

  > *"Not all prompts are available when downloading logs down, so it is only a certain newest cutoff available."*
  >
  > The task explicitly requires: *"complete logs of all project-related interactions with each system (prompts and outputs)... undeclared AI-assisted code or text is not permitted."* (pp. 1, 7)
  >
  > Admitting in the submitted text that your logs are incomplete is a direct compliance violation. You should format the declaration using the required template, export or reconstruct the full prompt histories from both authors, and ensure the `genai-logs/` directory contains all interactions without stating that logs were cut off.

## 2. Missing Core Task Requirements

- **Week 38: Inspection of High- and Low-Perplexity Examples:**

  Section 4 of the task states:

  > *"Inspect examples with high and low perplexity and formulate one hypothesis about what each model captures."*
  >
  > In Section 4.1 (p. 3), only aggregate perplexities are reported (1748.00, 81.63, 46.11, 78.62) with a general remark on trigram sparsity. The report **does not inspect or quote any specific validation question examples** with high or low perplexities, nor does it formulate qualitative hypotheses based on individual examples.

- **Week 37: Language Breakdown for Argmax and BIO Repair (Figure 4):**

  Section 1 mandates:

  > *"Report results both overall and separately for Arabic, Korean and Telugu, always with the number of evaluated examples."*
  >
  > In Figure 4 (p. 3), `Always O` and `Beam (β = 4)` report per-language rows (`ar`, `ko`, `te`, `All`), but `Indep. Argmax` and `BIO Repair` only report the aggregate `All` row ($N = 1,155$).

- **Weeks 40–41: Missing Answerability Class Metrics in Figure 5:**

  The task requires comparing integrated QA abstention decisions with the Week 39 answerability classifiers. In Figure 5 (p. 4), Week 36 and Week 39 report the full suite (`Ans-P`, `Ans-R`, `Ans-F1`, `Unans-P`, `Unans-R`, `Unans-F1`), but the W40–41 rows replace these columns with `SQuAD-v2 EM` and `SQuAD-v2 F1`. Consequently, there is no unanswerable precision/recall reported for XLM-RoBERTa to evaluate whether it resolved the severe unanswerable recall drop seen in Telugu during Week 39.

## 3. Evident Typos and Structural Refinements

- **Figure 8 Caption Duplicate Typo:**

  The caption for Figure 8 (p. 5) is an accidental duplicate of Figure 7:

  > *“Figure 8: Week 43 extensions compared against the frozen core XLM-RoBERTa QA model ($\tau^ = -5.1981$).”*
  >
  > Figure 8 is the **Challenge Set Evaluation** table (`Ans OK`, `Unans OK`, `Pairs`, `Pair Acc`). The caption must be changed to describe the challenge set results.

- **Challenge Set Inclusion of Extension 2:**

  The prompt specifies to evaluate *"the integrated QA model and each relevant extension"* on the challenge set (p. 6). Figure 8 includes Extension 1 (`W43: Translate-Test`), but omits Extension 2 (Russian Zero-Shot). If Extension 2 was excluded because all challenge pairs were authored in Arabic, Korean, and Telugu, state this explicitly in the text or table note to show it was not overlooked.

- **Week-by-Week Section Headings:**

Section 9 (p. 6) specifies:

> *"Organise the report as a coherent investigation rather than one section per teaching week."*
>
> While the top-level sections follow a thematic structure, the subsection titles remain literally named *"Week 36: Rule-Based Baselines"*, *"Week 37: Structured BIO Span Prediction"*, *"Week 38: Arabic Language Modelling"*, *"Week 39: Learned Answerability Classifiers"*, and *"Individual Extensions (Week 43)"*. Removing the teaching week labels in favor of descriptive titles (e.g., *"Rule-Based Baselines"*, *"BIO Sequence Tagging"*, *"Question Language Models"*) aligns directly with the grading rubric.

### 4. Final Submission Checklist

Ensure your submission ZIP matches the guidelines before the **30 October 17:00 Copenhagen time** deadline:

1. **Digital Exam Only:** Submit via Digital Exam; the brief specifies **not** to submit the final report on Absalon (p. 2).
2. **ZIP Contents:**
   - Validated `challenge.json` (verified with `python validate_challenge.py`).
   - Complete, runnable source code and environment specifications.
   - `genai-logs/` folder containing plain-text or PDF transcripts of all GenAI interactions.

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*User prompt: What about the structure the report, does it make sense how we have structured it?*

Response: The overall macro-structure of the report makes sense and closely aligns with standard empirical NLP papers, but it contains **three notable structural issues** and **one explicit rubric violation** regarding how sections are labeled and grouped.

### 1. The Explicit Rubric Violation: Syllabus-Based Sectioning

The assignment brief specifically instructs:

> *"Organise the report as a coherent investigation rather than one section per teaching week."*

While the top-level section headers (`3 Core Methods`, `4 Validation Results`, `5 Individual Extensions`) attempt a thematic framing, the sub-structure reverts to a literal weekly homework log:

- `3.1 Week 36: Rule-Based Baselines`
- `3.2 Week 37: Structured BIO Span Prediction`
- `3.3 Week 38: Arabic Language Modelling`
- `3.4 Week 39: Learned Answerability Classifiers`
- `3.5 Weeks 40–41: Integrated Transformer QA`
- In Section 4: `Linear BIO Bottleneck (Week 37)`, `Syntactic Abstention Priors (Week 39)`, `Joint XLM-RoBERTa (Weeks 40–41)`, `Week 39 BiLSTM Errors`, `Weeks 40–41 Transformer Errors`
- `5 Individual Extensions (Week 43)`

Graders scanning headings will immediately notice that the narrative follows the teaching calendar rather than an evolving scientific investigation.

**Fix:** Strip out every instance of "Week XX" from section titles, paragraph bolding, and table labels. Replace them with functional descriptions:

- `3.1 Heuristic & Surface Baselines` (instead of *Week 36*)
- `3.2 Token-Level BIO Sequence Tagging` (instead of *Week 37*)
- `3.3 Source Query N-gram Modeling` (instead of *Week 38*)
- `3.4 Discriminative Answerability Classification` (instead of *Week 39*)
- `3.5 End-to-End Multilingual Transformer QA` (instead of *Weeks 40–41*)

### 2. Fractured Error Analysis and Extension Flow

Section 9 of the brief suggests the following logical progression:

> *"research question and data; methods; validation results; challenge-set and error analysis; extensions and limitations; conclusion."*

In the current draft:

1. Validation error taxonomies are placed in **Section 4.3** (page 4).
2. Individual extensions are placed in **Section 5** (pages 4–5).
3. Challenge set evaluation is placed in **Section 6** (page 5).

Because Section 6 is essentially an adversarial error analysis investigating failure modes (entity salience vs. relational predicate checking), having the two Individual Extensions wedged between validation error analysis and the challenge set fragments the diagnostic storyline. Furthermore, Figure 8 evaluates Extension 1 on the challenge set, which creates forward- and backward-referencing between Sections 5 and 6.

**Two cleaner structural options:**

- **Option A (Follow the brief's suggested flow):**

  Group all diagnostic and error investigations together before extensions:

  `§4 Validation Results` → `§5 Diagnostic & Challenge Set Error Analysis` (combining the validation error taxonomy and the 10-pair challenge set findings) → `§6 System Extensions` (Translate-Test & Zero-Shot Russian) → `§7 Limitations and Conclusion`.

- **Option B (Keep current section ordering, but unify the narrative):**

  Keep Extensions as Section 5, but title Section 6 **"Stress-Testing via Paired Challenge Sets"**. In Section 6, explicitly state that after validating both core models and extensions, the challenge set serves as the ultimate diagnostic test for abstention brittleness across all systems.

### 3. Integrating the Arabic Language Model (Week 38)

In the current text, Section 3.3 introduces Arabic N-grams, and Section 4.1 reports validation perplexity. It is then never mentioned again in the paper.

Within a paper whose stated objective is cross-lingual extractive QA and abstention (context → English, questions → Arabic/Korean/Telugu), an isolated Arabic language model feels disconnected from the pipeline.

**Structural remedy:**

Provide a one-sentence narrative bridge in Section 3.3 explaining *why* query language modeling is investigated (e.g., *"To quantify query regularity and evaluate whether n-gram priors capture interrogative structure in non-Latin scripts independently of context grounding, we train..."*). Then, in Section 4.1, explicitly connect the low bigram perplexity to the syntactic findings in Section 4.2 (e.g., noting that predictable question prefixes directly explain why linear classifiers easily identify factoid WH-words).

### 4. Back Matter Ordering and Missing Sections

The prompt states:

> *"References, a reproducibility appendix, the contribution statement and AI declaration do not count towards the limit... Put a concrete contribution statement last."*

The current back matter structure on Pages 6 and 7 is ordered as follows:

1. `Contribution Statement` (Top of Page 6)
2. `Appendix A: Reproducibility Appendix` (Page 6)
3. `Appendix B: Generative AI Declaration` (Page 7)

This structure has two issues:

1. **References are completely missing**. References must be placed after the main body.
2. The Contribution Statement is placed *before* the appendices rather than **last**.

**Recommended back matter layout (starting on Page 6):**

- `References` (Unnumbered, immediately following Section 7)
- `Appendix A: Reproducibility Details` (Hardware, seeds, exact hyperparameters)
- `Appendix B: Generative AI Declaration` (Using the official KU template)
- `Contribution Statement` (Concrete itemized breakdown, placed as the final section of the document)

### Summary Checklist for Restructuring

| Current Structure | Issue | Recommended Revision |
|---|---|---|
| **Section 3 & 4 Subheadings** | Syllabus labels ("Week 36", "Week 37") violate Section 9. | Rename to conceptual titles (`Heuristic Baselines`, `BIO Sequence Tagging`, `Answerability Classification`, etc.). |
| **Section 4.3 vs. Section 6** | Error analysis is split into two disjoint sections across the paper. | Unify under a dedicated evaluation/error analysis progression or frame §6 as an adversarial stress-test. |
| **Arabic LM (§3.3 & §4.1)** | Isolated island; does not feed into QA or abstention narrative. | Add explicit transition linking question n-gram predictability to interrogative syntax exploitation. |
| **Back Matter Order** | No bibliography; contribution statement is not last. | **Order:** `References` → `Reproducibility Appendix` → `GenAI Declaration` → `Contribution Statement`. |

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User prompt: We need a challenge.json file. Following is from our notebook: import json from collections import Counter, defaultdict from pathlib import Path import re from typing import Any, Mapping, Sequence import pandas as pd # Complete Paired Challenge Set challenge_records = [ # VINCENT VARGA (5 PAIRS: CSGO / League of legends across ar, ko, te) { "id": "pair1_csgo_major_ans", "pair_id": "pair1_csgo_major", "created_by": "Vincent Varga", "question": ( "ما هو الفريق الذي فاز ببطولة دريمهك الشتوية لعام 2013 في لعبة كاونتر سترايك؟" ), "context": ( "DreamHack Winter 2013 was the first Counter-Strike: Global" " Offensive Major Championship. Fnatic won the tournament," " upsetting Ninjas in Pyjamas in the grand final." ), "lang": "ar", "answerable": True, "answer_start": 89, "answer": "Fnatic", "source_url": "https://en.wikipedia.org/wiki/DreamHack_Winter_2013", }, { "id": "pair1_csgo_major_unans", "pair_id": "pair1_csgo_major", "created_by": "Vincent Varga", "question": ( "ما هو الفريق الذي فاز ببطولة دريمهك الصيفية لعام 2013 في لعبة كاونتر سترايك؟" ), "context": ( "DreamHack Winter 2013 was the first Counter-Strike: Global" " Offensive Major Championship. 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The map was originally designed by David" " Johnston for the original Counter-Strike game." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/Dust_II", }, { "id": "pair3_lol_worlds_ans", "pair_id": "pair3_lol_worlds", "created_by": "Vincent Varga", "question": ( "లీగ్ ఆఫ్ లెజెండ్స్ మొదటి సీజన్ వరల్డ్ ఛాంపియన్‌షిప్‌ను ఏ జట్టు" " గెలుచుకుంది?" ), "context": ( "The Season 1 World Championship was held in June 2011 at DreamHack" " Summer in Sweden. Fnatic defeated against Against All Authority" " in the grand final to become the inaugural champions." ), "lang": "te", "answerable": True, "answer_start": 85, "answer": "Fnatic", "source_url": ( "https://en.wikipedia.org/wiki/League_of_Legends_World_Championship" ), }, { "id": "pair3_lol_worlds_unans", "pair_id": "pair3_lol_worlds", "created_by": "Vincent Varga", "question": ( "లీగ్ ఆఫ్ లెజెండ్స్ రెండవ సీజన్ వరల్డ్ ఛాంపియన్‌షిప్‌ను ఏ జట్టు" " గెలుచుకుంది?" ), "context": ( "The Season 1 World Championship was held in June 2011 at DreamHack" " Summer in Sweden. 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Slaying it rewards the entire" " surviving team with the Hand of Baron buff." ), "lang": "ko", "answerable": True, "answer_start": 22, "answer": "Baron Nashor", "source_url": "https://en.wikipedia.org/wiki/League_of_Legends", }, { "id": "pair5_lol_baron_unans", "pair_id": "pair5_lol_baron", "created_by": "Vincent Varga", "question": ( "리그 오브 레전드의 칼바람 나락에서 처치 시 내셔 남작의 손길 버프를" " 주는 몬스터는 무엇인가?" ), "context": ( "In League of Legends, Baron Nashor is the most powerful neutral" " monster on Summoner's Rift. 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*"lang": "ko", "answerable": True, "answer_start": 83, "answer": "15", "source_url": "https://en.wikipedia.org/wiki/UEFA_Champions_League", }, { "id": "bt_pair5_unans", "pair_id": "bt_pair5", "created_by": "Bertan Turan", "question": ( "레알 마드리드는 챔피언스리그 결승에서 몇 번 패배했는가?" ), "context": ( "Real Madrid is the most successful club in the tournament's" " history, having won it 15 times." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/UEFA_Champions_League", }, ] can you please make the syntax match for a dedicated .json file.*

Response: I will now generate the dedicated `challenge.json` file containing all 20 records (10 matched pairs) properly formatted as a JSON array according to the project specifications.

I have generated `challenge.json` with the exact JSON array schema required by the course guidelines. All string concatenations have been resolved, Python booleans converted to standard JSON `true`/`false`, and character substring offsets verified against the context strings.

You can save this file directly into the root of your project directory and run `python validate_challenge.py` to confirm verification passes before adding it to your submission ZIP.

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*User prompt: We need a challenge.json file. Following is from our notebook: import json from collections import Counter, defaultdict from pathlib import Path import re from typing import Any, Mapping, Sequence import pandas as pd # Complete Paired Challenge Set challenge_records = [ # VINCENT VARGA (5 PAIRS: CSGO / League of legends across ar, ko, te) { "id": "pair1_csgo_major_ans", "pair_id": "pair1_csgo_major", "created_by": "Vincent Varga", "question": ( "ما هو الفريق الذي فاز ببطولة دريمهك الشتوية لعام 2013 في لعبة كاونتر سترايك؟" ), "context": ( "DreamHack Winter 2013 was the first Counter-Strike: Global" " Offensive Major Championship. Fnatic won the tournament," " upsetting Ninjas in Pyjamas in the grand final." ), "lang": "ar", "answerable": True, "answer_start": 89, "answer": "Fnatic", "source_url": "https://en.wikipedia.org/wiki/DreamHack_Winter_2013", }, { "id": "pair1_csgo_major_unans", "pair_id": "pair1_csgo_major", "created_by": "Vincent Varga", "question": ( "ما هو الفريق الذي فاز ببطولة دريمهك الصيفية لعام 2013 في لعبة كاونتر سترايك؟" ), "context": ( "DreamHack Winter 2013 was the first Counter-Strike: Global" " Offensive Major Championship. Fnatic won the tournament," " upsetting Ninjas in Pyjamas in the grand final." ), "lang": "ar", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/DreamHack_Winter_2013", }, { "id": "pair2_csgo_dust2_ans", "pair_id": "pair2_csgo_dust2", "created_by": "Vincent Varga", "question": ( "카운터 스트라이크의 더스트 II 맵을 처음 설계한 사람은 누구인가?" ), "context": ( "Dust II is a video game map featured in the first-person shooter" " series Counter-Strike. The map was originally designed by David" " Johnston for the original Counter-Strike game." ), "lang": "ko", "answerable": True, "answer_start": 123, "answer": "David Johnston", "source_url": "https://en.wikipedia.org/wiki/Dust_II", }, { "id": "pair2_csgo_dust2_unans", "pair_id": "pair2_csgo_dust2", "created_by": "Vincent Varga", "question": ( "카운터 스트라이크의 더스트 II 맵의 그래픽 리메이크를 총괄한 사람은" " 누구인가?" ), "context": ( "Dust II is a video game map featured in the first-person shooter" " series Counter-Strike. The map was originally designed by David" " Johnston for the original Counter-Strike game." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/Dust_II", }, { "id": "pair3_lol_worlds_ans", "pair_id": "pair3_lol_worlds", "created_by": "Vincent Varga", "question": ( "లీగ్ ఆఫ్ లెజెండ్స్ మొదటి సీజన్ వరల్డ్ ఛాంపియన్‌షిప్‌ను ఏ జట్టు" " గెలుచుకుంది?" ), "context": ( "The Season 1 World Championship was held in June 2011 at DreamHack" " Summer in Sweden. Fnatic defeated against Against All Authority" " in the grand final to become the inaugural champions." ), "lang": "te", "answerable": True, "answer_start": 85, "answer": "Fnatic", "source_url": ( "https://en.wikipedia.org/wiki/League_of_Legends_World_Championship" ), }, { "id": "pair3_lol_worlds_unans", "pair_id": "pair3_lol_worlds", "created_by": "Vincent Varga", "question": ( "లీగ్ ఆఫ్ లెజెండ్స్ రెండవ సీజన్ వరల్డ్ ఛాంపియన్‌షిప్‌ను ఏ జట్టు" " గెలుచుకుంది?" ), "context": ( "The Season 1 World Championship was held in June 2011 at DreamHack" " Summer in Sweden. Fnatic defeated against Against All Authority" " in the grand final to become the inaugural champions." ), "lang": "te", "answerable": False, "answer_start": -1, "answer": "", "source_url": ( "https://en.wikipedia.org/wiki/League_of_Legends_World_Championship" ), }, { "id": "pair4_csgo_modes_ans", "pair_id": "pair4_csgo_modes", "created_by": "Vincent Varga", "question": ( "في لعبة كاونتر سترايك، ما هي الأداة التي يحاول الإرهابيون زرعها في المواقع المحددة؟" " " ), "context": ( "In the primary competitive game mode of Counter-Strike: Global" " Offensive, the Terrorists attempt to plant a C4 explosive at one" " of two designated bomb sites." ), "lang": "ar", "answerable": True, "answer_start": 108, "answer": "C4 explosive", "source_url": ( "https://en.wikipedia.org/wiki/Counter-Strike:_Global_Offensive" ), }, { "id": "pair4_csgo_modes_unans", "pair_id": "pair4_csgo_modes", "created_by": "Vincent Varga", "question": ( "في لعبة كاونتر سترايك، ما هي الأداة التي يحاول مكافحو الإرهاب زرعها في المواقع المحددة؟" " " ), "context": ( "In the primary competitive game mode of Counter-Strike: Global" " Offensive, the Terrorists attempt to plant a C4 explosive at one" " of two designated bomb sites." ), "lang": "ar", "answerable": False, "answer_start": -1, "answer": "", "source_url": ( "https://en.wikipedia.org/wiki/Counter-Strike:_Global_Offensive" ), }, { "id": "pair5_lol_baron_ans", "pair_id": "pair5_lol_baron", "created_by": "Vincent Varga", "question": ( "리그 오브 레전드의 소환사의 협곡에서 처치 시 내셔 남작의 손길" " 버프를 주는 몬스터는 무엇인가?" ), "context": ( "In League of Legends, Baron Nashor is the most powerful neutral" " monster on Summoner's Rift. Slaying it rewards the entire" " surviving team with the Hand of Baron buff." ), "lang": "ko", "answerable": True, "answer_start": 22, "answer": "Baron Nashor", "source_url": "https://en.wikipedia.org/wiki/League_of_Legends", }, { "id": "pair5_lol_baron_unans", "pair_id": "pair5_lol_baron", "created_by": "Vincent Varga", "question": ( "리그 오브 레전드의 칼바람 나락에서 처치 시 내셔 남작의 손길 버프를" " 주는 몬스터는 무엇인가?" ), "context": ( "In League of Legends, Baron Nashor is the most powerful neutral" " monster on Summoner's Rift. Slaying it rewards the entire" " surviving team with the Hand of Baron buff." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/League_of_Legends", }, # BERTAN TURAN (5 PAIRS: Football Domain in Korean 'ko') { "id": "bt_pair1_ans", "pair_id": "bt_pair1", "created_by": "Bertan Turan", "question": "리오넬 메시는 언제 태어났는가?", "context": ( "Lionel Andrés Messi (born 24 June 1987) is an Argentine" " professional footballer who plays as a forward for and captains" " Major League Soccer club Inter Miami." ), "lang": "ko", "answerable": True, "answer_start": 26, "answer": "24 June 1987", "source_url": "https://en.wikipedia.org/wiki/Lionel_Messi", }, { "id": "bt_pair1_unans", "pair_id": "bt_pair1", "created_by": "Bertan Turan", "question": "리오넬 메시는 언제 결혼했는가?", "context": ( "Lionel Andrés Messi (born 24 June 1987) is an Argentine" " professional footballer who plays as a forward for and captains" " Major League Soccer club Inter Miami." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/Lionel_Messi", }, { "id": "bt_pair2_ans", "pair_id": "bt_pair2", "created_by": "Bertan Turan", "question": ( "크리스티아누 호날두가 활약하고 있는 사우디 프로리그 구단은 어디인가?" ), "context": ( "Cristiano Ronaldo plays as a forward for Saudi Pro League club Al" " Nassr." ), "lang": "ko", "answerable": True, "answer_start": 63, "answer": "Al Nassr", "source_url": "https://en.wikipedia.org/wiki/Cristiano_Ronaldo", }, { "id": "bt_pair2_unans", "pair_id": "bt_pair2", "created_by": "Bertan Turan", "question": ( "크리스티아누 호날두가 소유하고 있는 사우디 프로리그 구단은 어디인가?" ), "context": ( "Cristiano Ronaldo plays as a forward for Saudi Pro League club Al" " Nassr." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url":*

*"https://en.wikipedia.org/wiki/Cristiano_Ronaldo", }, { "id": "bt_pair3_ans", "pair_id": "bt_pair3", "created_by": "Bertan Turan", "question": "2022년 FIFA 월드컵에서 우승한 국가는 어디인가?", "context": ( "The 2022 FIFA World Cup was won by Argentina, who defeated the" " defending champions France in the final." ), "lang": "ko", "answerable": True, "answer_start": 35, "answer": "Argentina", "source_url": "https://en.wikipedia.org/wiki/2022_FIFA_World_Cup", }, { "id": "bt_pair3_unans", "pair_id": "bt_pair3", "created_by": "Bertan Turan", "question": "2022년 FIFA 월드컵을 개최한 국가는 어디인가?", "context": ( "The 2022 FIFA World Cup was won by Argentina, who defeated the" " defending champions France in the final." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/2022_FIFA_World_Cup", }, { "id": "bt_pair4_ans", "pair_id": "bt_pair4", "created_by": "Bertan Turan", "question": ( "엘링 홀란드가 2022년 7월에 입단한 프리미어리그 구단은 어디인가?" ), "context": ( "Erling Haaland joined Premier League club Manchester City in July" " 2022." ), "lang": "ko", "answerable": True, "answer_start": 42, "answer": "Manchester City", "source_url": "https://en.wikipedia.org/wiki/Erling_Haaland", }, { "id": "bt_pair4_unans", "pair_id": "bt_pair4", "created_by": "Bertan Turan", "question": "엘링 홀란드가 2022년 7월에 떠난 프리미어리그 구단은 어디인가?", "context": ( "Erling Haaland joined Premier League club Manchester City in July" " 2022." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/Erling_Haaland", }, { "id": "bt_pair5_ans", "pair_id": "bt_pair5", "created_by": "Bertan Turan", "question": "레알 마드리드는 챔피언스리그에서 몇 번 우승했는가?", "context": ( "Real Madrid is the most successful club in the tournament's" " history, having won it 15 times." ), "lang": "ko", "answerable": True, "answer_start": 83, "answer": "15", "source_url": "https://en.wikipedia.org/wiki/UEFA_Champions_League", }, { "id": "bt_pair5_unans", "pair_id": "bt_pair5", "created_by": "Bertan Turan", "question": ( "레알 마드리드는 챔피언스리그 결승에서 몇 번 패배했는가?" ), "context": ( "Real Madrid is the most successful club in the tournament's" " history, having won it 15 times." ), "lang": "ko", "answerable": False, "answer_start": -1, "answer": "", "source_url": "https://en.wikipedia.org/wiki/UEFA_Champions_League", }, ] can you please make the syntax match for a dedicated .json file.*

Response: I will now format and generate the standalone `challenge.json` file, concatenating the multi-line question and context strings into standard JSON syntax and preserving all character offsets and required schema keys.

The Python dictionary records have been converted into a valid, formatted JSON array in `challenge.json`. All 20 items satisfy the course schema:

- `id` and `pair_id` are consistently mapped.
- Every answerable item has verified zero-based character offsets (`answer_start`) and matching substring spans (`answer`).
- Every unanswerable counterpart is initialized with `answer_start: -1` and `answer: ""`.
- Matched pairs share identical English contexts, creator names, languages, and source URLs.

You can save this directly into your repository root and confirm verification by running `python validate_challenge.py`.

---

*User prompt: Okay, can you help me do the references for the report please, we have not done them.*

### Response: 1. Ready-to-Use BibTeX File (`references.bib`)

Save the following BibTeX entries into a `references.bib` file in your LaTeX project root:

```
@inproceedings{clark2020tydi,
  title     = {{TyDi QA}: A Benchmark for Information-Seeking Question Answering in Typologically Diverse
  author    = {Clark, Jonathan H. and Choi, Eunsol and Collins, Michael and Garrette, Dan and Kwiatkowski,
  booktitle = {Transactions of the Association for Computational Linguistics (TACL)},
  volume    = {8},
  pages     = {454--470},
  year      = {2020}
}

@inproceedings{asai2021xor,
  title     = {{XOR QA}: Cross-lingual Open-Retrieval Question Answering},
  author    = {Asai, Akari and Kasai, Jungo and Clark, Jonathan H. and Lee, Kenton and Choi, Eunsol and Ha
  booktitle = {Proceedings of the 2021 Conference of the North American Chapter of the Association for Com
  pages     = {547--564},
  year      = {2021}
}

@inproceedings{muller2023evaluating,
  title     = {Evaluating and Modeling Attribution for Cross-Lingual Question Answering},
  author    = {Muller, Benjamin and Wieting, John and Clark, Jonathan H. and Kwiatkowski, Tom and Ruder, S
  booktitle = {Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMN
  pages     = {144--157},
  year      = {2023}
}

@inproceedings{rajpurkar2018know,
  title     = {Know What You Don't Know: Unanswerable Questions for {SQuAD}},
  author    = {Rajpurkar, Pranav and Jia, Robin and Liang, Percy},
  booktitle = {Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (AC
  pages     = {784--789},
  year      = {2018}
}

@inproceedings{conneau2020unsupervised,
```

```
  title     = {Unsupervised Cross-lingual Representation Learning at Scale},
  author    = {Conneau, Alexis and Khandelwal, Kartikay and Goyal, Naman and Chaudhary, Vishrav and Wenzek
  booktitle = {Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (AC
  pages     = {8440--8451},
  year      = {2020}
}

@article{nllb2022,
  title     = {No Language Left Behind: Scaling Human-Centered Machine Translation},
  author    = {{NLLB Team} and Costa-juss{\`a}, Marta R. and Cross, James and {\c{C}}elebi, Onur and Elbay
  journal   = {arXiv preprint arXiv:2207.04672},
  year      = {2022}
}

@inproceedings{kudo2018sentencepiece,
  title     = {{SentencePiece}: A simple and language independent subword tokenizer and detokenizer for Ne
  author    = {Kudo, Taku and Richardson, John},
  booktitle = {Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMN
  pages     = {66--71},
  year      = {2018}
}

@inproceedings{ramshaw1995text,
  title     = {Text Chunking using Transformation-Based Learning},
  author    = {Ramshaw, Lance A. and Marcus, Mitchell P.},
  booktitle = {Third Workshop on Very Large Corpora},
  pages     = {82--94},
  year      = {1995}
}

@article{hochreiter1997long,
  title     = {Long Short-Term Memory},
  author    = {Hochreiter, Sepp and Schmidhuber, J{\"u}rgen},
  journal   = {Neural Computation},
  volume    = {9},
  number    = {8},
  pages     = {1735--1780},
  year      = {1997}
}

@article{chen1999empirical,
  title     = {An Empirical Study of Smoothing Techniques for Language Modeling},
  author    = {Chen, Stanley F. and Goodman, Joshua},
  journal   = {Computer Speech \& Language},
  volume    = {13},
  number    = {4},
  pages     = {359--393},
  year      = {1999}
}

@inproceedings{loshchilov2017decoupled,
  title     = {Decoupled Weight Decay Regularization},
  author    = {Loshchilov, Ilya and Hutter, Frank},
  booktitle = {International Conference on Learning Representations (ICLR)},
  year      = {2019}
}

@inproceedings{wolf2020transformers,
  title     = {Transformers: State-of-the-Art Natural Language Processing},
  author    = {Wolf, Thomas and Debut, Lysandre and Sanh, Victor and Chaumond, Julien and Delangue, Clemen
  booktitle = {Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMN
  pages     = {38--45},
  year      = {2020}
}

@article{pedregosa2011scikit,
  title     = {Scikit-learn: Machine Learning in {P}ython},
  author    = {Pedregosa, Fabian and Varoquaux, Ga{\"e}l and Gramfort, Alexandre and Michel, Vincent and T
  journal   = {Journal of Machine Learning Research (JMLR)},
  volume    = {12},
  pages     = {2825--2830},
  year      = {2011}
}

@inproceedings{paszke2019pytorch,
  title     = {{PyTorch}: An Imperative Style, High-Performance Deep Learning Library},
  author    = {Paszke, Adam and Gross, Sam and Massa, Francisco and Lerer, Adam and Bradbury, James and Ch
  booktitle = {Advances in Neural Information Processing Systems (NeurIPS)},
  volume    = {32},
```

```
  year      = {2019}
}
```

### 2. In-Text Citation Placement Guide

Insert these citation commands into your report text to cover every benchmark, model, and tool mentioned:

- **Section 1: Introduction and Shared Protocol (Page 1)**
  - *Original sentence:* "We investigate this central question on the (coastalcph/tydi_xor_rc) dataset, focusing on Arabic (ar), Korean (ko), and Telugu (te)..."

    *Change to:*

    ```
    ...on the \texttt{coastalcph/tydi\_xor\_rc} dataset \citep{asai2021xor, clark2020tydi, muller2023evaluating}, focusing on Arabic...
    ```

  - *Original sentence:* "For integrated QA (Weeks 40–43), we report SQuAD-v2 normalized EM and token F1..."

    *Change to:*

    ```
    ...we report SQuAD-v2 normalized EM and token $F_1$ \citep{rajpurkar2018know} plus Answerability Macro-$F_1$...
    ```

- **Section 2: Tokenization & Subword Mechanics (Pages 1 & 2)**
  - *Original sentence:* "Figure 1: Dataset statistics by split and language using the xlm-roberta-base SentencePiece tokenizer."

    *Change to:*

    ```
    ...using the \texttt{xlm-roberta-base} SentencePiece tokenizer \citep{kudo2018sentencepiece, conneau2020unsupervised}.
    ```

- **Section 3.2: Structured BIO Span Prediction (Page 2)**
  - *Original sentence:* "We cast span extraction as token-level sequence labelling over English contexts tokenized via Unicode regex (\w+|[^\w\s]), mapping character offsets to {O, B-ANS, I-ANS}."

    *Change to:*

    ```
    ...mapping character offsets to \{O, B-ANS, I-ANS\} \citep{ramshaw1995text}.
    ```

  - *Original sentence:* "We train a linear SGDClassifier (log-loss, α = 10^-4, balanced class weights, seed 110)..."

    *Change to:*

    ```
    We train a linear SGDClassifier \citep{pedregosa2011scikit} ...
    ```

- **Section 3.3: Arabic Language Modelling (Page 2)**
  - *Original sentence:* "Against a Uniform Baseline (P = 1/|V|), our group compared three smoothed models..."

    *Change to:*

    ```
    ...our group compared three add-$\alpha$ smoothed models \citep{chen1999empirical}: Add-1.0 Unigram...
    ```

- **Section 3.4: Learned Answerability Classifiers (Page 3)**
  - *Original sentence:* "A single-layer bidirectional LSTM (h = 64) encodes Q (t = 32) and C (t = 256)..."

    *Change to:*

    ```
    A single-layer bidirectional LSTM \citep{hochreiter1997long} ($h = 64$) encodes...
    ```

- **Section 3.5: Integrated Transformer QA (Page 3)**
  - *Original sentence:* "We fine-tune a single joint multilingual Transformer (xlm-roberta-base) across Arabic, Korean, and Telugu..."

    *Change to:*

    ```
    We fine-tune a single joint multilingual Transformer (\texttt{xlm-roberta-base}) \citep{conneau2020unsupervised} across Arabic...
    ```

  - In the Reproducibility Appendix when referring to AdamW:

    ```
    ...optimized via AdamW \citep{loshchilov2017decoupled} ($lr = 3 \times 10^{-5}$)...
    ```

- **Section 5.1: Translate-Test QA Pipeline (Page 4)**
  - *Original sentence:* "Questions were translated into English via facebook/nllb-200-distilled-600M and evaluated through the frozen Weeks 40–41 XLM-RoBERTa model..."

    *Change to:*

```
Questions were translated into English via \texttt{facebook/nllb-200-distilled-600M}
\citep{nllb2022} and evaluated...
```

- **Appendix A.1: Hardware and Software Environment (Page 6)**
  - *Original sentence:* "Software versions recorded from the execution environment: Python 3.13, transformers5.20.2, datasets4.8.5, evaluate0.4.6, scikit-learn1.6.1, pandas2.2.3, numpy2.1.3, and torch..."

    *Change to:*

    ```
    ...transformers \citep{wolf2020transformers}, scikit-learn \citep{pedregosa2011scikit}, and
    PyTorch \citep{paszke2019pytorch}.
    ```

### 3. LaTeX Document Placement

In your main `.tex` file, place the bibliography commands directly at the end of Section 7 (before `\appendix`), ensuring the references appear right after the conclusion and do not count against the 5-page limit:

```
% End of Section 7 (Limitations and Conclusion) on Page 5

\bibliographystyle{acl_natbib}
\bibliography{references}

\appendix

\section{Reproducibility Appendix}
% Hardware, seeds, and hyperparameters...

\section{Generative AI Declaration}
% Official KU template...

\section*{Contribution Statement}
% Concrete individual contributions placed LAST as required by Section 9
```

---